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| UNIVERSITE DE BOUMERDES<br>UNIVERSITY OF BOUMERDES | *IGEE*<br>*University M'Hamed*<br>*Bougara of Boumerdès* | ***Last Name:*** .....<br>***First Name:*** .....<br>***Group:*** ..... | ***Grade :*** |

***EE321: Computer Architecture***
***Control Exam (1h30)***
***2021 – 2022***

**Notes: Answer briefly and clearly using the provided space. No extra sheet will be accepted.**

## 1. Basic Concepts (8 pts)

1.1 Consider a CPU running at 2 GHz with 200 instructions, 100 instructions are executed in 4 clock cycles, 60 instructions in 7 clock cycles and 40 instructions in 10 clock cycles. What is the speed (MIPS) of the CPU?

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1.2 Give the functional table of an active low 3-state buffer.

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1.3 Give one application of the 3-state buffer.

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1.4 Consider 2's complement representation using 20 bits, complete the following equalities:

$(+15)_{10}$ = (.....)$_{2'C}$ $(-15)_{10}$ = (.....)$_{2'C}$

$(00F00H)_{2'C}$ = (.....)$_{10}$ $(FF100H)_{2'C}$ = (.....)$_{10}$

1.5 Consider a 16-bit CPU and the HEX operation: 00F0H-00FFH. Which of the flags: carry, overflow, sign, and zero are set by this operation. Justify.

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1.6 Give the main advantage and disadvantage of the CMOS technology compared to NMOS technology.

1.7 Draw the transistor schematic of the CMOS gate that implements: $Y = \overline{\overline{ABCD}.\overline{ABC}}$.

## 2. CPU Function and Program Execution (12 pts)

Consider a Little-Endian simple 8-bit CPU with the following specifications: 8-bit address bus, memory is byte addressable, instructions opcodes are given in Table 1, binary codes of the general purpose registers (GPR) are given in Table 2, and the instructions formats are given in Table 3 and Table 4. After the fetch cycle of the first instruction of the program, we have: PC=96H, MBR=D0H, and IR=D080H, and after the fetch cycle of the last instruction, we have PC=B6H, MBR=F8H and IR=E0F8H.

2.1 Give the size of the registers : PC, MAR, IR, and MBR. Justify.

2.2 Give the instructions that are using indirect access memory.

2.3 Give the HEX code of the first and last instruction. Justify

2.4 Give the content of the registers R0, PC, MAR, MBR, and IR after the execute cycle of the first instruction.

2.5 Give the content of the registers R0, PC, MAR, MBR, and IR after the execute cycle of the last instruction.

2.6 What is the size of the program? Justify.

2.7 Using register R0 and R1, write an optimized assembly program to add the value -8 to the content of memory location 80H.

2.8 Give the HEX code and the size of your program.

| Instruction | Opcode (hex) | Description |
|---|---|---|
| MV Rd, X | 1H | Rd = X |
| MV Rd, Rs | 2H | Rd = Rs |
| LD Rd, Rs | 3H | Rd = [Rs] |
| LD Rd, [X] | 4H | Rd = [X] |
| LD Rd, # Rs | 5H | Rd =[ [Rs]] |
| ST Rd, X | 6H | [Rd] = X |
| ST Rd, Rs | 7H | [Rd] = Rs |
| AD Rd, X | 8H | Rd = Rd+X |
| AD Rd, Rs | 9H | Rd = Rd+Rs |
| AD Rd, [Rs] | AH | Rd = Rd+[Rs] |
| AD Rd, [X] | BH | Rd =Rd+ [X] |
| AD Rd, # X | CH | Rd=Rd+[[X]] |
| XOR Rd, Rs | DH | Rd=Rd xor Rs |
| JP X | EH | PC = PC+X<br>X= 8-bit 2's complement displacement |
| HL | FH | Halt |

*Rd = Source/destination register (it can be any GPR).*
*Rs = Source register (it can be any GPR)*
*X = 8-bit value/address*
*R0 = Register R0*
*[X] = content of memory location X.*

**Table 1: Instructions description.**

| Register | Code |
|---|---|
| R0 | 00 |
| R1 | 01 |
| R2 | 10 |
| R3 | 11 |

**Table 2: Binary codes of GPRs.**

| 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Opcode | | | | 0 | 0 | Rd | | Operand reference X | | | | | | | |

**Table 3: 2-Bytes instructions format (for the instructions using operand reference X).**

| 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|---|---|
| Opcode | | | | Destination register (Rd) | | Source register (Rs) | |

**Table 4: 1-Byte instructions format (instructions without the operand reference X).**